

Shivam Deolankar

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SUMMARY

A **systems programmer** with **3 years** of experience working on **low-level systems** in **C and Rust**, focusing **performance** and **memory safety**. An active **open-source contributor**, proficient in **backend**, **OS internals**, **embedded systems**, and **debugging**.

EDUCATION

ABV - Indian Institute of Information Technology and Management, Gwalior Aug 2023 – Apr 2028
Integrated B.Tech. + M.Tech. in Information Technology **CGPA - 7.95/10 (6th Sem)**

EXPERIENCE

ML Research Intern @ ABV-IIITM Gwalior | May 2026 – Aug 2026

Adaptive Wavelet Attention for improving Energy Forecasting Models

- Developed a **Band Attention module for SimpleTM**, inspired by **FEDformer's frequency domain approach**.
- Enabled **adaptive frequency weighting** through **wavelet bands** to capture multi-scale patterns in electricity load dataset.
- Improved **forecasting accuracy** across **5+ horizons**, achieving **23.5% higher R²**, **17.4% higher EVS**, and **1.3% lower MSE**.

RTEMS - Real Time Operating System | Jan 2026 – Jun 2026

Open Source Contributor

- Resolved a **Stack Checker Reporter defect** by restructuring flag handling and decoupling **SECTION FLAGS** from **COMPILER FLAGS**, eliminating flag conflicts and ensuring consistent configuration, improving overall **build reliability**.
- Resolved **POSIX B0 baud rate edge case** issue across UART drivers for **13 BSPs and drivers**, implementing runtime safeguard mechanisms to prevent **divide-by-zero**, and standardizing **termios handling** for robust serial communication.
- Implemented **fallback-safe MMU mapping** for missing **PCIe nodes** in **device trees**, ensuring reliable handling of incomplete hardware descriptions and enabling stable **boot** with correct memory initialization across BSP environments.
- Migrated FatFS codebase**, fixed **error.h macro conflicts** and **libdl overflow**, improving build stability and safety.

PROJECTS

SMTP Server | Aug 2026 – Aug 2026

Custom SMTP Server using Golang (smtpd), Node.js (smtp-server), AWS EC2, Cloudflare DNS

- Implemented a custom **AWS EC2 instance** to securely handle incoming emails using a self-managed infrastructure.
- Integrated the instance with a custom domain via **Cloudflare DNS** by setting up isolated MX records, effectively masking the personal email address and enforcing strict receive-only protocols for enhanced security and privacy.

PikaNote | Feb 2025 – Aug 2025

A light weight terminal text editor in Rust featuring efficient rendering and low-level control.

- Implemented **rich text editing features** including **insertion, deletion, line breaks, cursor navigation, and file I/O**.
- Built **real-time terminal interaction** using Crossterm, handling **key events** and **dynamic cursor control** responsively.
- Implemented **incremental screen rendering** using queued terminal updates to minimize redraw overhead efficiently.
- Developed **bidirectional incremental search** with **live highlighting** using an annotated string system for large buffers.
- Designed a **modular terminal UI** with **3 components** including **editor view, message/command bar** and a **status bar**.

Embedded Systems Projects | Jan 2024 – Apr 2026

Real-time hardware systems with sensors and embedded ML.

- Built an **avionics telemetry tool** processing **100+ records** and **15+ anomalies** with validation, filtering, and reporting.
- Developed a gesture-controlled **Glove Mouse** using ESP32 and MPU6050, mapping gyroscope data to real-time cursor control.
- Built an **Arduino-based rotten fruit segregation system** using **ML classification** with **servo-driven actuation**.

ACHIEVEMENTS

- Overall Winner of Webkriti 2024**, organised by **AASF** for developing Yatramitra, a travel ticket booking platform.
- Secured **3rd position** in Product Management Track of **AASF Winter Projects, 2025**.

POSITIONS OF RESPONSIBILITY

- Core Team Lead**, Alumni Meet 2026, **ABV-IIITM Gwalior**.
- Technical and Design Lead**, IEEE Student Branch, **ABV-IIITM Gwalior**.
- Executive**, Student Activity Council 2025 (Technical), at **ABV-IIITM Gwalior**.
- Facilitated a 2-day workshop on **Web Basics - JS** for **50+ students** at **ABV-IIITM Gwalior**.
- Designed official cover page for **Abhishar V15**, the annual magazine of **ABV-IIITM Gwalior** published by **AASF**.

TECHNICAL SKILLS

Systems Programming	Rust, Golang, JavaScript, C++, C, Python
Backend Frameworks	Axum, Tokio, Reqwest, Node.js, Express.js, Redis, Kafka, REST APIs, Websockets
Databases and ORM	SQL, PostgreSQL, MongoDB, Diesel, Sqlx, SurrealDB
DevOps	Docker, CI/CD, Amazon Web Services, AWS Lambda, EC2, S3
AI and ML	MCP, Vector Embeddings, Transformers, LLMS, Predictive Load Forecasting
Development Tools	Git, GitHub, Gitlab, Claude Code, Arch Linux, Postman, HTML, TailwindCSS, React
Other Tools	ABI, RTOS, GDB, QEMU, CMake, Canva, Photoshop, Figma, Notion